

ing its cause. It fails to answer the question of what it is that computers do—or what it is that people do with computers—that causes educated workers to be relatively more in demand.

This paper proposes an answer to this question. We formalize and test a simple theory of how the rapid adoption of computer technology—spurred by precipitous real price declines—changes the tasks performed by workers at their jobs and ultimately the demand for human skills. Our approach builds on an intuitive set of observations offered by organizational theorists, computer scientists, and most recently economists about what computers do—that is, the tasks they are best suited to accomplish—and how these capabilities complement or substitute for human skills in workplace settings.² The simple observations that undergird our analysis are (1) that computer capital substitutes for workers in carrying out a limited and well-defined set of cognitive and manual activities, those that can be accomplished by following explicit rules (what we term “routine tasks”); and (2) that computer capital complements workers in carrying out problem-solving and complex communication activities (“nonroutine” tasks). (See Table I for examples.) Provided that routine and nonroutine tasks are imperfect substitutes, these observations imply measurable changes in the task composition of jobs, which we test below.

To answer the core questions of our paper, the ideal experiment would provide two identical autarkic economies, one facing a dramatic decline in the price of computing power and the other not. By contrasting these economies, it would be straightforward to assess how computerization reshapes the task composition of work and hence the structure of labor demand. Because this experiment is not available, we develop a simple economic model to predict how demand for workplace tasks responds to an economywide decline in the price of computer capital. The model predicts that industries and occupations that are initially intensive in labor input of routine tasks will make relatively larger investments in computer capital as its price declines. These industries and occupations will reduce labor input of routine tasks, for which computer

2. Simon [1960] provides the first treatment of this question with which we are familiar, and his essay introduces many of the ideas explored here. Other early works include Drucker [1954] and Nelson and Winter [1982]. Adler [1986], Orr [1996], and Zuboff [1988] discuss what computers and related technology do in the workplace, but do not consider economic implications. Acemoglu [1998], Goldin and Katz [1998], Bresnahan [1999], Bartel, Ichniowski, and Shaw [2000], Lindbeck and Snower [2000], Lang [2002], and Bresnahan, Brynjolfsson, and Hitt [2002] provide economic analyses of why technology and human capital are complementary.

capital substitutes, and increase demand for nonroutine task input, which computer capital complements. In net, these forces will raise relative demand for highly educated workers, who hold comparative advantage in nonroutine versus routine tasks.

To test these predictions, we pair representative data on job task requirements from the *Dictionary of Occupational Titles (DOT)* with samples of employed workers from the Census and Current Population Survey to form a consistent panel of industry and occupational task input over the four-decade period from 1960 to 1998. A unique virtue of this database is that it permits us to analyze changes in task input within industries, education groups, and occupations—phenomena that are normally unobservable. By measuring the tasks performed in jobs rather than the educational credentials of workers performing those jobs, we believe our study supplies a missing conceptual and empirical link in the economic literature on technical change and skill demand.

Our analysis provides four main pieces of evidence supporting our model.

- 1) Commencing in the 1970s, labor input of routine cognitive and manual tasks in the U. S. economy declined, and labor input of nonroutine analytic and interactive tasks rose.
- 2) Shifts in labor input favoring nonroutine and against routine tasks were concentrated in rapidly computerizing industries. These shifts were small and insignificant in the precomputer decade of the 1960s, and accelerated in each subsequent decade.
- 3) The substitution away from routine and toward nonroutine labor input was not primarily accounted for by educational upgrading; rather, task shifts are pervasive at all educational levels.
- 4) Paralleling the within-industry task shifts, occupations undergoing rapid computerization reduced input of routine cognitive tasks and increased input of nonroutine cognitive tasks.

We consider a number of economic and purely mechanical alternative explanations for our results. Two supply side factors that we study in particular are the rising educational attainment of the workforce and the rising human capital and labor force attachment of women—both of which could potentially generate shifts in job task composition independent of demand shifts. As we show below, the task shifts that we document—and their

associations with the adoption of computer technology—are as pervasive within gender, education, and occupation groups as between, indicating that these supply side forces are not the primary explanation for our results.

We begin by presenting our informal “task model” describing how computerization affects the tasks that workers and machines perform. We next formalize this model in a production framework to develop empirical implications for task demand at the industry and occupation level. Subsequent sections describe our data sources and test the model’s main implications. Drawing together the empirical strands, we finally assess the extent to which changes in the task composition can account for recent demand shifts favoring more educated workers. This exercise shows that estimated task shifts are economically large, underscoring the potential of the conceptual model to reconcile key facts.

I. THE TASK MODEL

We begin by conceptualizing work from a “machine’s-eye” view as a series of tasks to be performed, such as moving an object, performing a calculation, communicating a piece of information, or resolving a discrepancy. Our model asks: which of these tasks can be performed by a computer? A general answer is found by examining what is arguably the first digital computer, the Jacquard Loom of 1801. Jacquard’s invention was a machine for weaving fabrics with inlaid patterns specified by a program punched onto cards and fed into the loom. Some programs were quite sophisticated; one surviving example uses more than 10,000 cards to weave a black and white silk portrait of Jacquard himself.³ Two centuries later, the electronic descendants of Jacquard’s loom share with it two intrinsic traits. First, they rapidly and accurately perform repetitive tasks that are deterministically specified by stored instructions (programs) that designate unambiguously what actions the machine will perform at each contingency to achieve the desired result. Second, computers are “symbolic processors,” acting upon abstract representations of information such as binary numbers or, in the loom’s case, punched cards.

Spurred by a more than trillionfold decline in the real price of

3. The Jacquard loom was also the inspiration for Charles Babbage’s analytical engine and Herman Hollerith’s punch card reader, used to process the 1910 United States Census.

VII. CONCLUSION

What is it that computers do—or what is it that people do with computers—that appears to increase demand for educated workers? This paper formalizes and tests an intuitive answer to this question that has been informally articulated by scholars in a number of disciplines over several decades. Computer technology substitutes for workers in performing routine tasks that can be readily described with programmed rules, while complementing workers in executing nonroutine tasks demanding flexibility, creativity, generalized problem-solving capabilities, and complex communications. As the price of computer capital fell precipitously in recent decades, these two mechanisms—substitution and complementarity—have raised relative demand for workers who hold a comparative advantage in nonroutine tasks, typically college-educated workers.

Our task framework emphasizes that the causal force by which advancing computer technology affects skill demand is the declining price of computer capital—an economywide phenomenon. We developed a simple model to explore how this price decline alters task demand within industries and occupations. This model predicts that industries that were intensive in labor input of routine tasks in the precomputer era would make relatively larger investments in computer capital. Simultaneously, they would reduce labor input of routine tasks, for which computer capital substitutes, and increase demand for nonroutine task input, which computer capital complements.

Employing consistent, representative, time series observations on the task composition of jobs from the *Dictionary of Occupational Titles*, we affirm these predictions across several margins of task change and estimate that they may have contributed substantially to demand shifts favoring educated labor over the past three decades. We also considered several alternative explanations for our findings, most significantly the rising human capital and labor force attachment of women. We find that the documented task shifts, and their associations with the adoption of computer technology, are as evident *within* gender, education, and occupation groups as between them. The pervasiveness of these shifts suggests to us that changes in job task content—spurred by technological change—may plausibly be viewed as an underlying factor contributing to recent demand shifts favoring educated labor.